

Financial News Sentiment Prediction

using Deep Learning & BERT

1. Objective

Predict the sentiment of financial news text as **Bullish**, **Bearish** or **Neutral**.

2. Dataset

Twitter Financial News Sentiment Dataset

3. Models Used

LSTM

Recurrent model, reads text in sequence

BERT

Transformer model, learns context

4. Data Preprocessing

- Removed URLs
- Removed mentions
- Tokenization

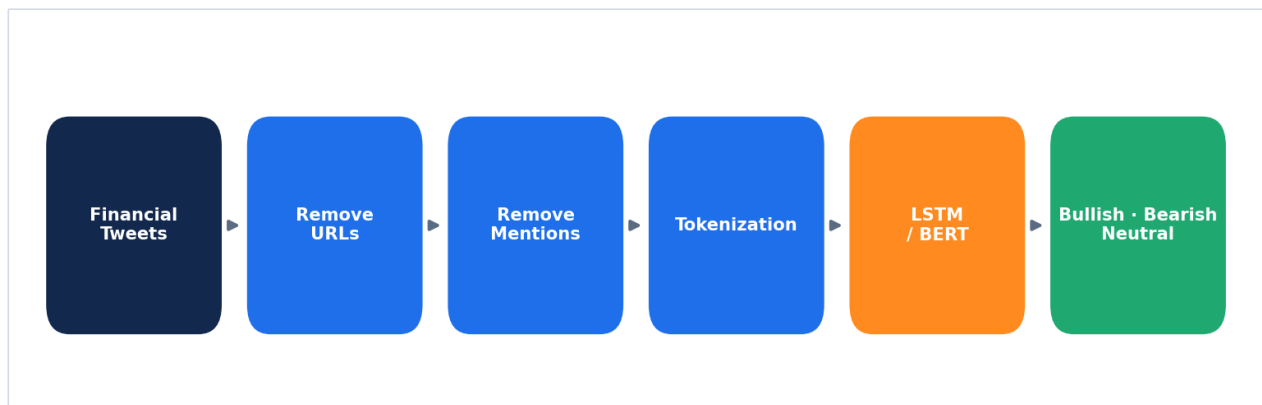


Figure 1 Project pipeline

5. Results

MODEL	ACCURACY	F1 SCORE
LSTM	82%	0.80
BERT	95%	0.94

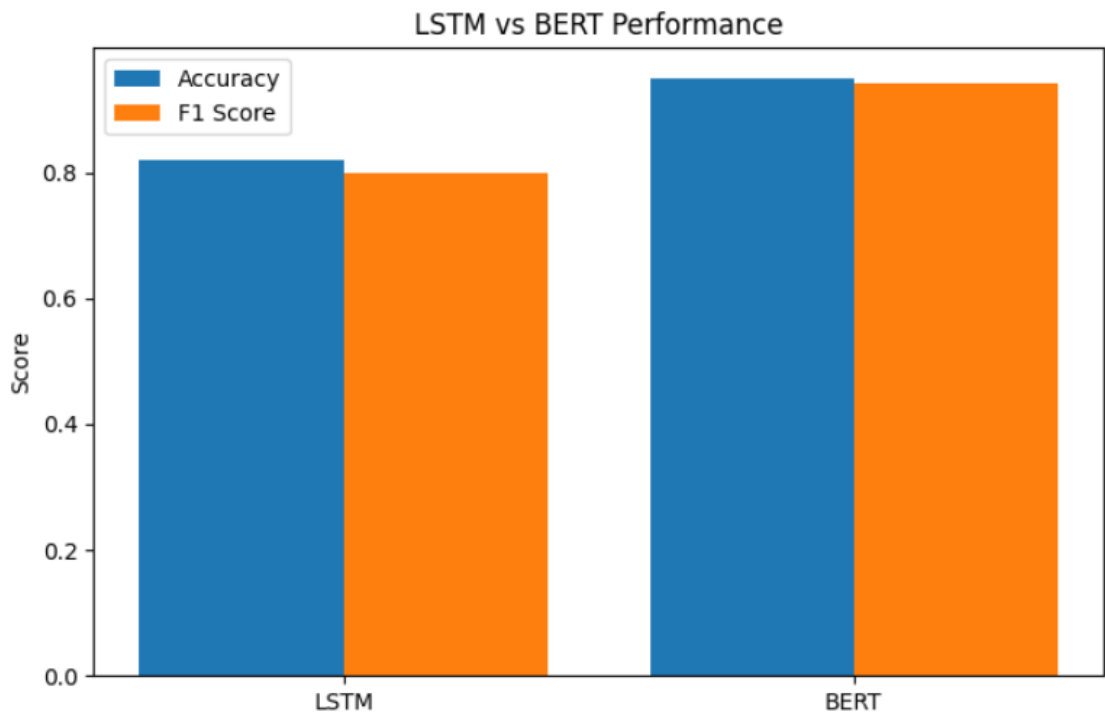


Figure 2 Accuracy and F1 score, LSTM vs BERT

6. Model Comparison

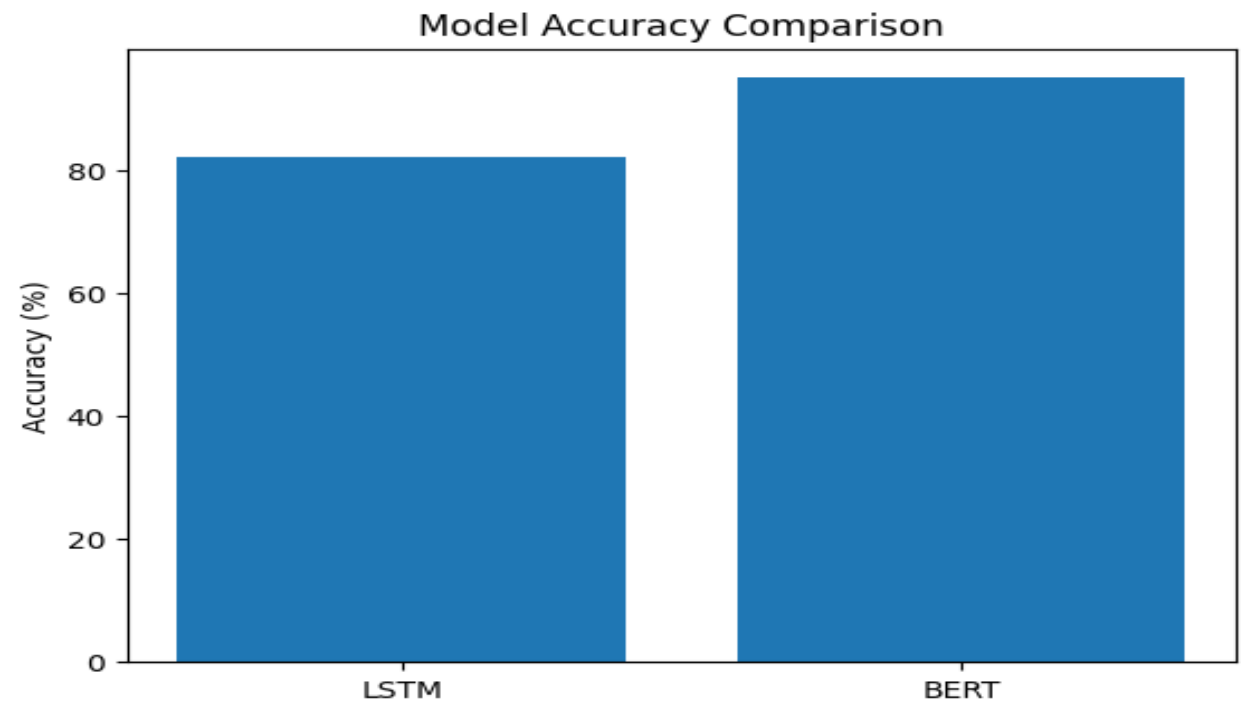


Figure 3 Accuracy comparison

Conclusion

BERT outperformed LSTM (95% vs 82% accuracy, 0.94 vs 0.80 F1) because it understands the contextual meaning of text.